

Jonathan Cruz

SENIOR BACKEND & PAYMENTS ENGINEER

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PROFILE

"Senior software engineer with over 10 years of experience building fintech and payments infrastructure. Deep expertise in transactional Postgres, double-entry ledgers, KYC/EDD compliance, and integrations with banking and identity providers (Plaid, Treasury Prime). Strong TypeScript across the stack with NestJS, Prisma, and event-driven architectures designed for fault tolerance, idempotency, and auditability."

PROFESSIONAL EXPERIENCE

SELECTED ROLES & CONTRIBUTIONS

FINTECH FOR VC FUND
ADMINISTRATION

Sydecar

FULL-STACK SOFTWARE
ENGINEER

Double-Entry
Ledger

POSTGRES

Contributed to a proprietary double-entry ledger engine (Fragment) with append-only immutable entries, deterministic idempotency keys (), derived balances (never stored as source of truth), and minor-unit integer accounting with banker's rounding. Posting flow is event-driven and fully reconstructable from source events.

POSTGRES LEDGER DOUBLE-ENTRY DECIMAL

Plaid +
Treasury
Prime

BANKING

Worked on Plaid Link integration (createLinkToken → exchangePublicToken → access_token) with processor tokens powering ACH transfers via Treasury Prime. Built a mock banking engine for the local sandbox with idempotency tracking by application/account IDs.

PLAID TREASURY PRIME ACH BANKING

KYC/EDD

COMPLIANCE

Designed and implemented Enhanced Due Diligence including LP funding blocks, UBO verification, FinCEN integration, CTA compliance, and automatic KYC revalidation for active investors. Built race-resistant verify-identity flow with re-check inside a to handle "email-free-at-blur → taken-at-submit" edge cases (409 EMAIL_NOW_EXISTS).

COMPLIANCE KYC UBO FINCEN

Webhook
Idempotency

CONCURRENCY

Implemented webhook dedup via (unique) with PROCESSING/PROCESSED/REJECTED state machine, and an API-level idempotency interceptor with 48h TTL scoped by .

IDEMPOTENCY WEBHOOKS CONCURRENCY

Fault
Tolerance

BULLMQ OUTBOX

Standardized event delivery using BullMQ Pro with exponential backoff (5 retries / 3s base in ledger; 20 retries / 10s base in pubsub). Outbox pattern: persist resolution first, enqueue after — jobs are idempotent and survive enqueue failure. Global per-deal ordering via Redis INCR counters with deterministic predecessor lookup ().

BULLMQ OUTBOX REDIS FAULT-TOLERANCE

Billing V2

ARCHITECTURE

Designed (ADR) and implemented fund billing V2 with quarterly fee scheduler, billing model versioning, and administrative fee structure management.

TS-REST ZOD NESTJS

QuickBooks

INTEGRATION

Built bidirectional invoice sync engine with mock service for local dev, state validation, auto-sync on approval, and full E2E test suite with Playwright and Cucumber BDD.

QUICKBOOKS API PLAYWRIGHT CUCUMBER

ts-rest

TYPE-SAFE
CONTRACTS

Migrated legacy controllers (auth, onboarding, admin) to type-safe ts-rest contracts with Zod schemas, improving type safety end-to-end.

TYPESCRIPT ZOD TS-REST

MAY 2023 — PRESENT
SAN FRANCISCO, CA (REMOTE)

B2B PAYMENTS & WHOLESALE

Zax

SR SOFTWARE ENGINEER

JAN 2019 — MAY 2023
SÃO PAULO

B2B Payments

MOBILE

Built from scratch a wholesale ordering app with integrated payment flows and credit-limit validation for business partners, using React Native, TypeScript, and GraphQL.

REACT NATIVE

TYPESCRIPT

GRAPHQL

Serverless

AWS

Deployed full cloud infrastructure on AWS (EC2, ECS, S3, Lambda, VPCs) with serverless web apps using React and Next.js.

AWS

NEXT.JS

SERVERLESS

CI/CD

AUTOMATION

Configured end-to-end delivery pipeline with GitHub Actions, Fastlane, and Code Climate for quality control. Docker for local dev.

GITHUB ACTIONS

FASTLANE

DOCKER

CRYPTO CONSUMER FINTECH

Netlolo

SR SOFTWARE ENGINEER

SEP 2018 — JAN 2019
SÃO PAULO

Crypto Wallet

WEB3

Built a mobile + web app bringing cryptocurrency to mainstream users with React Native and TypeScript, integrating Web3 smart contracts on Ethereum for self-custodial wallets and on-chain transactions.

REACT NATIVE

TYPESCRIPT

WEB3

ETHEREUM

Chat SDK

REAL-TIME

Developed a chat SDK for businesses using React Native and React/Redux, powered by XMPP and a Java backend for real-time messaging.

REACT

REDUX

XMPP

JAVA

Zero

DOWNTIME

Built CI/CD automations ensuring continuous deployment with no service interruption. PostgreSQL with Hasura as the GraphQL layer.

POSTGRESQL

HASURA

CI/CD

CONSUMER HEALTHCARE

Escale

SR SOFTWARE ENGINEER

JAN 2018 — SEP 2018
SÃO PAULO

Healthcare Platform

REACT

Built the product foundation with React, Redux, and a Node.js/TypeScript backend for healthcare services sold via Google Ads, including lead capture, eligibility checks, and provider matching.

REACT

REDUX

NODE.JS

TYPESCRIPT

Mentorship

TECH TALKS

Led internal Tech Talks on RESTful design, Functional Programming, and Microservices architecture. MySQL, PostgreSQL, and MongoDB depending on service responsibility.

MICROSERVICES

CIRCLECI

MONGODB

STUDENT LOANS FINTECH

PraValer

PL SOFTWARE ENGINEER

JUN 2017 — JAN 2018
SÃO PAULO

Loan Origination

NODE

Migrated legacy PHP loan origination platform to a Node.js/Express backend for a student loans product with governmental funding, handling loan request lifecycle, eligibility, and disbursement flows.

NODE.JS

EXPRESS

LOANS

PHP

Database

OPTIMIZATION

Restructured PostgreSQL transactional tables with improved schemas and indexes for loan ledger entries. MongoDB for loan requests and logs. Frontend with React and Redux.

POSTGRESQL

MONGODB

REACT

REDUX

CORE SKILLS

PAYMENTS & FINTECH

Plaid, Treasury Prime, ACH, KYC/EDD, FinCEN, Double-entry ledgers, Idempotency, Webhooks

BACKEND

TypeScript, NestJS, Node.js, Prisma, PostgreSQL, ts-rest, Zod, GraphQL

RELIABILITY

BullMQ, Outbox pattern, Event ordering, Redis, Datadog

FRONTEND

React, Next.js, React Native, TanStack Query

INFRASTRUCTURE

AWS, GCP, Docker, CI/CD, Nx Monorepo

LANGUAGES

PORTUGUESE

Native

ENGLISH

C1/C2

EDUCATION

Anhanguera

BACHELOR OF TECHNOLOGY (B.TECH.) IN INFORMATION TECHNOLOGY ·
JAN 2016 — JUL 2018